Kayshav Bhardwaj

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Education

Purdue University **Expected Graduation - May 2028**
Bachelor of Science in Artificial Intelligence *West Lafayette, Indiana*
Mission San Jose High School **Class of 2024**
Fremont, California

Relevant Coursework

- CS 373 - Data Mining and Machine Learning
- CS 243 - Artificial Intelligence Basics
- CS 176 - Data Engineering In Python
- STAT 511 - Statistical Methods
- PHIL 256 - Philosophy of Artificial Intelligence
- CS 439 - Data Visualization

Experience

ACTION National Science Foundation Artificial Intelligence Institute - *Research Fellow* **Jun 2026 – Present**

- Researching token entanglement in subliminal learning within LLMs under Dr. Yigitcan Kaya at UC Santa Barbara
- Designed a multi-stage pipeline (logit probes, ablations, cross-model attacks, prompt ladders) across 14+ conditions to characterize when and how LLM instruction secrets become recoverable.

Purdue AI For Musicians Lab Co-Lead - *Undergraduate Research Assistant* **Aug 2024 - Present**

- Co-authored a paper accepted at **NeurIPS 2025** AI for Music Workshop detailing competition results and benchmarking of AMT models from Google, Spotify, ByteDance, and competitors.
- Led over 50 members in the research team developing algorithms for polyphonic automatic music transcription (AMT).
- Organized the IEEE-sponsored 2025 AMT Challenge, attracting international participation.

Berkeley AI Research (BAIR) Lab - *Student Researcher* **Jun 2023 – Present**

- Research project with a team of UC Berkeley AI Research Lab (BAIR) Researchers Under Prof. Kurt Keutzer Using Machine Learning to Translate Underserved Asian Languages to English Text.
- Developed MITRA-Swarm, an agent-swarm machine translation framework, significantly outperforming standard foundation model baselines in translation quality.
- Engineered a dual-judge evaluation framework and conducted a 12-condition ablation across 1,000 text samples.

Purdue Governance and Responsible AI Lab (GRAIL) - *Dammon Dean's Fellow* **Mar 2025 - Present**

- Responsible AI Research under Professor Schiff in the Purdue GRAIL Lab.
- Contributing to the creation of high quality datasets of scholarly knowledge about AI's workforce impact.

The Data Mine of the Rockies - *Undergraduate Researcher* **Aug 2025 – May 2026**

- Collaborated with the National Space Intelligence Center (NSIC) and KBR on GPS signal anomaly detection.
- Trained a machine learning model to detect SNR anomalies in time series GNSS data to be used by the US Space Force.

Publications

LLMs Leak Secrets Through Unrelated Outputs ***NeurIPS 2026 InfPriv Workshop***

- First author of a NeurIPS 2026 workshop paper studies the privacy-preserving ability of LLMs under a threat model.
- Determined how LLMs can statistically leak secret PINs through unrelated outputs without explicitly leaking them.

MITRA-Swarm ***NeurIPS 2026 LP4FM Workshop***

- Lead author of NeurIPS 2026 workshop paper designing a multi-agent architecture for Sanskrit translation.
- Agent swarm architecture outperforms compute-matched self-refinement baselines.

Advancing Multi-Instrument Music Transcription: Results from the 2025 AMT Challenge | ***arXiv:2603.27528***

- Co-authored NeurIPS 2025 workshop paper benchmarking global research teams against SOTA transcription models.
- Analyzed AMT failure modes, identifying critical weaknesses in instrument leakage and melody disentanglement.

Technical Skills

Languages: Python, Java, Swift, SQL, R, Various AI Tools and APIs, Google Colab, GitHub, Google Suite, Microsoft Office
Other Skills: Leadership, Research, Communication, Teamwork, Initiative

Awards and Accomplishments

- 2025-2026 Computing Research Association Outstanding Undergraduate Researcher Award Honorable Mention
- Dean's List and Semester Honors (2024, 2025, 2026)
- Hello World Hackathon Organizer (2025)